

NFC Book I L4 Equivalence Certificate

Schema v0.37.0

L4 Spine Release Candidate

Generated by NFC Governance Engine v37

Contents

Book I L4 Certificate

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Book I is the unconditional source architecture. All theorem-like objects are [U]. This is an L4 spine-pilot shell — the canonical Book I is the authoritative source.

Book I L4 Equivalence Certificate

L4 gate: PASSED (9/9 criteria)

All results unconditional: True

K0=7 theorem: True

Defect ledger complete: True

Prose completeness: 100.0%

Book I is the unconditional source architecture. All theorem-like objects are [U] or [D]. No conditional theorems exist. The $K_0=7$ uniqueness theorem is the source of the gauge algebra identification inherited by the YM and SM branches.

L4 Criteria

Check	St.	Observed
l4.i.primary-exports-present	✓	all present
l4.i.all-results-unconditional	✓	all [U]
l4.i.k0-prime-present	✓	present=True, status=U
l4.i.no-presentation-surplus-present	✓	present
l4.i.source-arch-rules	✓	all present
l4.i.defect-ledger-complete	✓	complete
l4.i.prose-completeness	✓	100.0% (22/22)
l4.i.xref-annotations	✓	22/22
l4.i.book-i-boundary	✓	none

Primary Unconditional Exports

Machine ID	Canonical	Description
thm:I.governing	thm:governing (Boo	Packages both main exports: quotient well-defined + fin
thm:I.stabilization	thm:stabilization	Every refinement chain on finite Omega stabilizes finit
cor:I.no-presentation-surplus	cor:no-presentatio	Primary downstream governance rule; cited by Book VII
thm:I.K0-prime	thm:K0-prime	Unique prime fixed point of $F(p)$; source of $su(3)+su(2)$
prop:I.defect-nonneg	prop:defect-nonneg	$D(\chi) \geq 0$; foundational for defect accounting
prop:I.ledger-additivity	prop:ledger-additi	$D(\chi_1 + \chi_2) = D(\chi_1) + D(\chi_2)$
cor:I.defect-welldef	cor:defect-welldef	Complete defect ledger characterization
thm:I.entropy-monotone	thm:entropy-monoto	H_n non-increasing; pairs with defect via duality
thm:I.ledger-entropy-duality	thm:ledger-entropy	$D(\chi) = H_0 - H_\infty$; information-theoretic interpretation

Book I Boundary

- Local stability of observable outcomes — belongs to Books II/III
- Boundary determinacy — belongs to Books II/IV
- Source universality — belongs to Book III
- Branch legitimacy — belongs to Book V
- Continuum correspondence — belongs to Book VI